1.2.1 Alternatives

There are two main alternative models to the data mesh's decentralization of responsibility for data. We discuss their setups in more detail in chapter 6, but provide a quick overview here.

The first option is the centralization of both people and technology. This is the default setup for any startup. And it's a decent default option, just as the monolith is a decent default option for any software component. In the beginning, the costs of decentralization outweigh its benefits. The benefits brought in by working closely together inside one data team, having just one technology to use, make things a lot easier.

KEY POINT Centralization is a sensible default option for both organizational and technical data work. Decentralization does carry costs, and centralization can mitigate those. That option does imply, though, that the value derived from centralized and decentralized data is roughly equal.

The second option splits up the work not by business domains, as data mesh suggests, but by technology. This usually results in one core data engineering team being responsible mostly for ingesting data and provisioning a data storage infrastructure. Multiple other teams (analytics teams, data science teams, analysts, you name it) pick up the raw data and turn it into something meaningful down the road. You might first centralize your data system and then layer up with this option to increase the flow.

There is nothing wrong with these two approaches. They might be reasonable default options, but both fail to align with value creation, which is deeply tied to business domains. Neither option is able to address sudden changes in just one business domain. As with microservices, which enable you to quickly extract value from one specific service by scaling it up all by itself, the data mesh is able to scale up value extraction in just one domain. All other options need to scale up everything in order to scale up value extraction in just one domain.

So, one way or another, both of these alternatives will hit a wall at some point: adding the next data source or data science project will feel extremely complex and costly compared to earlier ones. That's the point where you want to switch to a data mesh.